1. What Is AI Bias?

1.1 Understanding AI Bias

AI bias occurs when an artificial intelligence system learns or follows patterns that lead to unfair outcomes for certain individuals or groups.

To understand this concept, consider a company that has developed an AI system to screen job applicants and decide which candidates should be selected for interviews.

The company trains the AI system using historical recruitment data. By analysing this data, the AI learns patterns associated with candidates who were previously selected or rejected.

However, a problem may arise if the historical data contains unfair patterns. For example, if male candidates were selected more frequently than female candidates in the past, the AI system might learn an incorrect association between gender and suitability for a job.

As a result, the system might rank a female candidate lower than a male candidate, even when both candidates have the same skills, qualifications, and professional experience.

This is a simple example of AI bias: a situation in which an AI system produces potentially unfair outcomes because of patterns learned from data, the way the system is designed, or how it is used.

1.2 Practical Example of AI Bias in Recruitment

Consider two candidates applying for the same position:

Evaluation Criteria	Candidate A	Candidate B
Skills	Same	Same
Qualifications	Same	Same
Professional Experience	Same	Same
AI Evaluation Score	85/100	60/100

Suppose the only difference between the two candidates is their gender, and the AI system assigns them significantly different scores.

This difference should be investigated to determine whether gender, or a factor associated with gender, has influenced the AI system's decision.

However, different scores alone do not conclusively prove that the system is biased. Further investigation is required to understand the reasons behind the difference and determine whether the outcome is unfair.

1.3 Important Considerations

When evaluating AI bias, it is important to understand the following points:

1. **Different outcomes do not automatically indicate bias.** Differences in AI-generated scores may result from legitimate differences in the information provided or from other factors affecting the system's output.
2. **Historical data can introduce bias.** If the data used to train an AI system reflects past discrimination or unequal opportunities, the system may learn and reproduce those patterns.
3. **Similar candidates should be evaluated fairly.** When candidates have equivalent job-related qualifications and experience, investigators should examine whether irrelevant characteristics, such as gender, influence the system's evaluations.
4. **Further investigation is necessary.** A difference in scores is a warning sign that may justify an audit, but additional evidence is needed to establish whether the system produces discriminatory or otherwise unfair outcomes.

2. How Does AI Become Biased?

Understanding How AI Bias Develops

To understand how an AI system can become biased, let us consider a simple example.

Imagine that you are teaching mathematics to a student, but you only allow the student to practise addition problems. Later, when you ask the student to solve multiplication problems, they may struggle or provide incorrect answers because they have not received enough practice with multiplication.

A similar situation can occur with artificial intelligence. An AI system learns patterns from the data it receives. If its training data is incomplete, unrepresentative, or contains unfair patterns, the system may produce inaccurate or unfair results when dealing with new situations.

AI bias can develop for several reasons, including the quality of the data, the patterns learned by the model, the instructions given to the system, and the information retrieved by supporting technologies.

Major Reasons Behind AI Bias

Reason 1: Incorrect or Incomplete Data

AI systems depend on data to learn patterns and generate predictions. If the training data represents only a limited group of people or situations, the system may not perform equally well for everyone.

Example:

Suppose a company trains an AI hiring system using recruitment records containing mostly male candidates. The system receives very few examples of successful female candidates.

As a result, the system may not learn enough about the qualifications and experiences of successful female applicants. It could therefore evaluate some female candidates inaccurately.

Key point: Incomplete or unrepresentative data can prevent an AI system from working reliably across different groups of people.

Reason 2: Learning Incorrect or Unfair Patterns

Even when a dataset contains a large amount of information, it may include historical inequalities or discriminatory practices. An AI system can learn these patterns and reproduce them in its decisions.

Example:

Suppose a company historically promoted male employees more frequently than female employees, even when some female employees had equivalent qualifications and performance.

If an AI system is trained to predict promotions using these historical decisions, it may learn patterns that favour male employees.

The system may then recommend male employees for promotion more frequently, reproducing the unfairness present in the historical data.

Key point: More data does not automatically mean better or fairer AI. The quality and fairness of the data also matter.

Reason 3: Biased Prompts or Instructions

The instructions given to an AI system can influence its responses. If a prompt contains stereotypes, unfair assumptions, or subjective criteria, the system may produce results that reflect those assumptions.

Example:

Consider the following instruction given to an AI recruitment assistant:

"Identify candidates who appear suitable for a senior technology leadership role. Prefer candidates who demonstrate a traditionally masculine leadership style."

This instruction introduces a gender-related assumption into the evaluation process. The AI may consequently favour candidates who match that description rather than assessing all candidates using relevant professional criteria.

A more appropriate instruction would be:

"Evaluate candidates using job-related criteria, including technical expertise, leadership experience, communication skills, and demonstrated professional achievements. Apply the same criteria consistently to all candidates."

Key point: AI outputs depend not only on the model and its data but also on how the task is defined and what instructions the system receives.

Reason 4: Incorrect or Irrelevant Information Retrieved by a RAG System

Retrieval-Augmented Generation (RAG) is a technique that allows an AI system to retrieve information from external documents or databases before generating an answer.

However, a RAG system may retrieve incomplete, outdated, irrelevant, or inaccurate documents. If the AI relies on that information, its response may be incorrect or misleading.

Example:

Suppose a company develops an AI assistant that answers questions about its employee promotion policies. The system uses internal HR documents to generate answers.

A user asks:

"What qualifications are required for promotion to a senior position?"

However, the RAG system retrieves an outdated policy document that lists requirements from several years ago.

The AI may generate an answer based on the outdated policy, even though the company has since introduced new promotion criteria.

If the retrieved documents also contain unfair assumptions, those assumptions may influence the response.

Key point: A RAG system can produce unreliable or unfair answers when the information it retrieves is inaccurate, incomplete, or inappropriate for the question.

Where Can AI Bias Originate?

Source	How It Can Cause Problems
Training data	The data may be incomplete or unrepresentative of different groups.
Historical patterns	The model may learn and reproduce past discrimination or inequality.
Prompts and instructions	The system may be guided by stereotypes or unfair evaluation criteria.
Retrieved information in RAG	The system may rely on outdated, irrelevant, or inaccurate documents.

3. How Can We Determine Whether an AI System Is Biased?

One of the most important steps in an AI audit is determining whether an AI system produces biased or potentially unfair outcomes.

For example, suppose there is a concern that an AI-powered recruitment system unfairly rejects female candidates.

It would be inappropriate to conclude that the system is biased without supporting evidence. Instead, we should design controlled tests, collect results, and analyse whether the system consistently produces unfair differences.

There are several methods that can help identify potential AI bias. Three important approaches are counterfactual testing, statistical fairness testing, and language-based bias evaluation.

Counterfactual Testing — Changing One Variable at a Time

Counterfactual testing examines whether changing a particular characteristic of an input changes the output of an AI system while keeping other relevant characteristics constant.

Example: Testing an AI Recruitment System

Suppose we want to investigate whether an AI recruitment system evaluates male and female candidates differently.

We create two fictional candidate profiles.

Test A: Candidate Rahul

- Professional experience: 5 years
- Technical skills: Python, React, and Node.js
- Education: Same degree and qualifications as the other candidate

Test B: Candidate Riya

- Professional experience: 5 years
- Technical skills: Python, React, and Node.js
- Education: Same degree and qualifications as the other candidate

Assumption: Both profiles are identical in all relevant job-related aspects, except for the candidate's name, which is intended to act as a gender-associated signal.

We submit both profiles to the same AI recruitment system and compare the results.

Suppose the system produces the following scores:

Evaluation	Rahul	Riya
Experience	5 years	5 years
Technical skills	Identical	Identical
Qualifications	Identical	Identical
AI evaluation score	85/100	60/100

The difference in scores raises a concern that the name may influence the system's evaluation.

However, this result alone does not prove that the system is biased. It indicates a potential unfair difference that requires further investigation.

What Should We Do Next?

To investigate the issue more reliably, we should:

1. **Test multiple equivalent candidate pairs.** Use a sufficiently large set of fictional profiles rather than relying on a single example.
2. **Use different names and controlled variations.** Test multiple names associated with different genders while keeping job-related qualifications equivalent.
3. **Keep testing conditions consistent.** Use the same model version, settings, evaluation criteria, and input format wherever possible.
4. **Repeat the tests.** Determine whether the observed differences occur consistently or may be caused by random variation.
5. **Analyse the results.** Measure the size and frequency of the differences and investigate possible explanations.

This approach helps determine whether the system's outputs are sensitive to a characteristic that should not affect the job-related evaluation.

Important distinction: When testing only names, we are primarily testing sensitivity to gender-associated names, not necessarily gender itself. Names can also convey other information, so additional controlled tests may be necessary.

Statistical Fairness Testing — Comparing Multiple Results

A single test cannot establish whether an AI system consistently treats different groups unfairly. Statistical fairness testing examines patterns across a larger number of cases.

Example: Comparing Two Candidate Groups

Suppose we evaluate an AI recruitment system using two groups of 100 candidates each. The groups are designed to have equivalent job-related qualifications.

The following hypothetical results are observed:

Evaluation Metric	Group A	Group B
Total candidates evaluated	100	100
Candidates selected	70	45
Selection rate	70%	45%

This means that Group A has a selection rate 25 percentage points higher than Group B.

Such a difference is worth investigating, particularly if the groups have equivalent job-related qualifications.

However, the result does not automatically prove discrimination. Before reaching a conclusion, we must examine the following factors:

- Whether the two groups are genuinely comparable in relevant qualifications.
- Whether the sample size and sampling method are appropriate.
- Whether the difference is statistically reliable or could reasonably arise from random variation.
- Whether the selection criteria or other stages of the recruitment process contribute to the difference.
- Whether the observed difference represents unfair treatment under the applicable fairness criteria.

This approach is commonly described as statistical fairness testing. It helps identify disparities that require further investigation.

Important: Selection rates are only one way to evaluate fairness. Depending on the application, an audit may also need to examine false rejection rates, false acceptance rates, ranking differences, and whether the system performs equally well across groups.

Evaluating Language Bias in Large Language Models

Bias can also appear in the language generated by large language models (LLMs), such as ChatGPT and other AI assistants.

In this method, we provide equivalent prompts that differ only in a controlled characteristic and compare the generated responses.

Example: Testing Gender-Related Language Bias

Prompt A:

> Write a professional description of a male software engineer.

Prompt B:

> Write a professional description of a female software engineer.

We then evaluate the responses using the following criteria:

1. **Professionalism:** Are both individuals described with a similar level of professionalism and respect?
2. **Leadership qualities:** Does one response emphasise leadership and technical expertise while the other focuses primarily on supportive or administrative qualities?
3. **Stereotypical assumptions:** Does either response introduce unnecessary gender stereotypes?

4. **Level of detail:** Are both descriptions equally detailed and informative?
5. **Relevance:** Are the descriptions based on the requested professional role rather than assumptions associated with gender?

Interpreting the Results

Suppose the AI describes the male software engineer as technically skilled, innovative, and capable of leading a team, while describing the female software engineer primarily as helpful, cooperative, and supportive.

This difference may indicate that the model is associating certain professional qualities with gender.

However, one pair of responses is insufficient to establish a general pattern. The test should be repeated with multiple prompts, different contexts, and controlled variations.

Where possible, the evaluation should also account for the model's randomness, differences in wording, and the subjectivity involved in judging language.

If similar differences appear consistently across a well-designed set of tests, they can provide evidence of potential language bias and justify a more detailed investigation.

4. How Can AI Bias Be Reduced or Mitigated?

Once potential bias has been identified in an AI system, the next step is to determine its underlying cause and select an appropriate corrective measure.

An important principle of AI auditing is that **the solution should address the root cause of the problem rather than simply modifying the system's output.**

Consider a simple example: your motorcycle refuses to start. You would not immediately replace its engine without investigating the problem. Instead, you would check whether the battery is discharged, the fuel tank is empty, or another component is malfunctioning.

The same principle applies to AI systems. Bias may originate from training data, the model's learned behaviour, the instructions provided to it, or the information retrieved by supporting systems.

Therefore, different causes require different mitigation strategies.

1. Incorrect, Incomplete, or Unrepresentative Data

Problem: The training dataset does not adequately represent the population or situations in which the AI system will be used. It may also contain incorrect labels or historical inequalities.

Possible solution: Audit the dataset, correct data-quality issues, improve representation, and review potentially problematic labels.

Example

Suppose an AI recruitment system is trained using historical hiring data that contains very few examples of successful female candidates.

The system may not learn sufficiently from the experiences and qualifications of those candidates.

To address this problem, the audit team could:

1. Examine the dataset to identify missing or underrepresented groups.
2. Verify whether candidate records and selection labels are accurate.
3. Add suitable, representative examples where appropriate.
4. Review whether historical hiring decisions reflect unfair practices.
5. Retest the system on representative evaluation data to determine whether its performance and fairness have improved.

Adding more examples alone may not solve the problem if the new data contains the same historical biases. The quality, relevance, and labels of the additional data must also be assessed.

2. The Model Has Learned an Unfair Pattern

Problem: Even when the dataset has been reviewed, the model may learn associations that produce unfair outcomes.

Possible solution: Evaluate suitable model-level interventions, such as retraining, fine-tuning, or fairness-aware training techniques.

Example

Suppose a recruitment model consistently assigns lower scores to female candidates whose job-related qualifications are equivalent to those of male candidates.

The engineering team could investigate whether the model has learned an inappropriate association between gender-related signals and candidate suitability.

Possible corrective measures include:

1. Reviewing the model's training process and evaluation criteria.
2. Fine-tuning or retraining the model using suitable training examples.
3. Testing fairness-aware training techniques designed to reduce unjustified disparities.
4. Evaluating whether the model focuses on relevant job-related qualifications.
5. Comparing the model's performance and fairness metrics before and after the intervention.

The objective is to reduce unfair differences without unnecessarily damaging the model's ability to identify qualified candidates.

Retraining or fine-tuning is not automatically effective. Each intervention must be evaluated to establish whether it improves fairness while maintaining acceptable accuracy and reliability.

3. The Prompt or Instructions Introduce Unfair Assumptions

Problem: The instructions given to the AI contain stereotypes, subjective criteria, or assumptions that may influence its response.

Possible solution: Rewrite the instructions using clear, relevant, and consistently applicable evaluation criteria.

Example

An AI recruitment assistant receives the following instruction:

"Select candidates who demonstrate a traditionally masculine leadership style."

This instruction introduces a gender-associated assumption that may not be relevant to the job.

A more appropriate instruction would be:

"Evaluate candidates using job-related criteria, including technical skills, relevant experience, leadership achievements, and the stated requirements of the position. Apply the same criteria consistently to all candidates."

After updating the prompt, the team should:

1. Run the same controlled test cases used during the original audit.
2. Compare the new outputs with the original results.
3. Check whether the identified disparities have decreased.
4. Verify that the system still performs its intended task accurately.
5. Test additional scenarios to determine whether the improvement generalises beyond the original examples.

Prompt improvements can help reduce instruction-related bias, but they cannot guarantee that all bias originating from the model or its training data will disappear.

4. A RAG System Retrieves Incorrect or Irrelevant Documents

Problem: A Retrieval-Augmented Generation (RAG) system retrieves incomplete, outdated, irrelevant, or inaccurate information. The language model then uses that information to generate an answer.

Possible solution: Improve document quality, retrieval logic, ranking, and the overall information-retrieval process.

Example

Suppose an AI assistant answers questions about a company's recruitment policy using internal HR documents.

During an audit, the team discovers that the system frequently retrieves an outdated recruitment policy instead of the current version.

Changing the prompt alone may not solve the underlying issue because the AI is receiving the wrong source material.

The team should investigate the retrieval pipeline and consider the following measures:

1. **Document quality:** Check whether the source documents are accurate, complete, and up to date.
2. **Document indexing:** Verify that the correct documents and their contents have been indexed properly.
3. **Retrieval quality:** Test whether the search system retrieves documents relevant to the user's question.
4. **Ranking and filtering:** Improve the prioritisation of relevant and authoritative documents, including the handling of outdated versions.
5. **Source-grounded responses:** Encourage the model to base its answers on retrieved evidence and acknowledge when the available information is insufficient.
6. **Retesting:** Run the same evaluation questions again and compare retrieval accuracy and answer quality before and after the changes.

Improving retrieval quality may reduce incorrect answers, but the retrieved information must still be evaluated for potential bias. Accurate retrieval does not automatically guarantee fair or unbiased responses.

5. How Can We Determine Whether an AI Bias Mitigation Strategy Is Effective?

One of the most important stages of an AI bias audit is evaluating whether the corrective measures implemented have actually improved the system's fairness.

Identifying potential bias and applying a mitigation technique are not sufficient on their own. The system must be tested again to determine whether the intervention has reduced the identified problem without introducing other undesirable effects.

For example, suppose an initial audit reveals a selection-rate difference of 25 percentage points between two comparable candidate groups. The engineering team then modifies the training data or changes the AI system.

After implementing the intervention, the same evaluation is repeated, and the difference decreases to 5 percentage points.

This reduction is a positive indication that the intervention may have improved fairness. However, it does not automatically establish that the improvement is reliable or that the system is now free from bias.

Further evaluation is necessary to determine whether the observed improvement is meaningful, consistent, and sustainable.

Baseline and After-Mitigation Comparison

The process of comparing an AI system's performance before and after a mitigation strategy is implemented is commonly referred to as a *baseline versus post-mitigation evaluation*.

The baseline represents the system's performance before the intervention. The post-mitigation evaluation measures its performance after the intervention under comparable testing conditions.

What Must Be Verified After Mitigation?

A reliable evaluation should examine several factors.

1. Is the Improvement Statistically Reliable?

The observed reduction could partly result from random variation, particularly when the sample size is small or the AI system produces variable outputs.

To investigate this, the audit team should:

- Use an appropriate sample size and evaluation method.
- Repeat tests when the model or evaluation process is stochastic.
- Calculate suitable uncertainty estimates, such as confidence intervals.
- Use appropriate statistical tests to compare the baseline and post-mitigation results.

The choice of statistical method should reflect the study design, including whether the same candidates or independent candidate groups are being evaluated.

2. Has the Model's Accuracy Decreased?

A mitigation technique might reduce differences between groups while also making the system less effective at identifying qualified candidates.

For example, the selection-rate difference may decrease, but the model may also reject more qualified candidates from both groups.

Therefore, the audit team should compare relevant performance metrics before and after mitigation, such as:

- Accuracy, where appropriate.
- Precision and recall.
- False-positive and false-negative rates.
- Task-specific performance measures.

The appropriate metrics depend on the system's purpose and the consequences of its decisions.

3. Has a New Problem Appeared in Another Group?

An intervention may improve outcomes for one group while producing new disparities for another.

For example, a recruitment model may show a smaller gender-related selection-rate difference after mitigation but develop a larger disparity across age groups.

The audit should therefore evaluate all relevant groups and, where appropriate, intersections of characteristics rather than examining only the group that originally revealed the problem.

4. Does the Improvement Generalise to New Test Cases?

An AI system may perform better on the examples used during mitigation but fail on new examples.

To reduce this risk, the audit team should evaluate the system using a separate test dataset that was not used to develop or tune the mitigation strategy.

Additional testing should cover different names, candidate profiles, job roles, and realistic operating conditions where relevant.

5. Does the Improvement Persist Over Time?

AI systems and their operating environments can change. New training data, model updates, changes in user behaviour, or modifications to retrieval sources may affect fairness.

Where appropriate, organisations should establish ongoing monitoring procedures to detect whether previously corrected disparities reappear.

6. Conclusion

Evaluating an AI bias mitigation strategy requires more than observing a reduction in one fairness metric. The audit team must determine whether the improvement is statistically reliable, whether the system continues to perform its intended task effectively, whether other groups experience new disparities, and whether the improvement holds on previously unseen test cases.

A baseline versus post-mitigation comparison provides a structured way to assess these changes.

The objective is to establish, using reproducible evidence, whether the intervention has meaningfully reduced the identified unfairness without introducing unacceptable new problems.